1.5.2 Containments $\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N)$ give rise to

$$Y(N) \twoheadrightarrow Y_1(N) \twoheadrightarrow Y_0(N)$$

$$\Gamma(N)\tau \mapsto \Gamma_1(N)\tau \mapsto \Gamma_0(N)\tau .$$

Describe corresponding maps between moduli spaces.

Sol:

$$\begin{array}{ccccc} Y(N) & \twoheadrightarrow & Y_1(N) & \twoheadrightarrow & Y_0(N) \\ \wr\wr & & \wr\wr & & \wr\wr \\ S(N) & \twoheadrightarrow & S_1(N) & \twoheadrightarrow & S_0(N) \end{array}$$

$$[\mathbb{C}/\Lambda_\tau, (\tau/N+\Lambda_\tau, 1/N+\Lambda_\tau)] \mapsto [\mathbb{C}/\Lambda_\tau, 1/N+\Lambda_\tau] \mapsto [\mathbb{C}/\Lambda_\tau, \langle 1/N+\Lambda_\tau\rangle]$$

So in general $[E,(P,Q)] \mapsto [E,Q] \mapsto [E,\langle Q\rangle]$.

1.5.3. Since containments are normal $\Gamma(N) \triangleleft \Gamma_1(N) \triangleleft \Gamma_0(N)$, the quotient $\Gamma_1(N)/\Gamma(N) \simeq \mathbb{Z}/N\mathbb{Z} \curvearrowright Y(N)$

$$(n, \Gamma(N)\tau) \sim (\gamma_1+\Gamma(N), \Gamma(N)\tau) \mapsto \gamma_1\Gamma(N)\tau \sim \Gamma(N)\gamma_1(\tau).$$

$$\Gamma_0(N)/\Gamma_1(N) \simeq (\mathbb{Z}/N\mathbb{Z})^\times \curvearrowright Y_1(N).$$

$$(n, \Gamma_1(N)\tau) \sim (\gamma_0+\Gamma_1(N), \Gamma_1(N)\tau) \mapsto \gamma_0\Gamma_1(N)\tau \sim \Gamma_1(N)\gamma_0(\tau).$$

Describe corresponding actions on moduli spaces.

Sol: $\cancel{S_1(N)/S(N)} \simeq \Gamma_1(N)/\Gamma(N) \simeq \mathbb{Z}/N\mathbb{Z} \curvearrowright Y(N) \simeq S(N)$.

$$(n, \Gamma(N)\tau) \sim (\gamma_1+\Gamma(N), \Gamma(N)\tau) \mapsto \gamma_1\Gamma(N)\tau \sim \Gamma(N)\gamma_1(\tau)$$

$$\sim [\mathbb{C}/\Lambda_{\gamma_1(\tau)}, \gamma_1(\tau)/N+\Lambda_{\gamma_1(\tau)}, 1/N+\Lambda_{\gamma_1(\tau)}] \qquad [E,(P,Q)]$$

$$\overset{m=j(\gamma_1,\tau)}{\sim} [\mathbb{C}/\Lambda_\tau, (m\gamma_1(\tau)/N+\Lambda_\tau, m/N+\Lambda_\tau)] \qquad \downarrow m=1 \quad [E,(P+nQ,Q)]$$

$$(\mathbb{Z}/N\mathbb{Z})^\times \curvearrowright Y_1(N) \simeq S_1(N) \quad (d, \Gamma_1(N)\tau) \mapsto \Gamma_1(N)\gamma_0(\tau)$$

$$\sim [\mathbb{C}/\Lambda_{\gamma_0(\tau)}, 1/N+\Lambda_{\gamma_0(\tau)}] \overset{m=j(\gamma_0,\tau)}{\sim} [\mathbb{C}/\Lambda_\tau, m/N+\Lambda_\tau] \qquad [E,Q] \xrightarrow{\gamma_0=\begin{pmatrix} a & c \\ N & d\end{pmatrix}} [E, d^*Q]$$

1.5.4. Let $w_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix} \in GL_2^+(\mathbb{Q})$.

Show w_N normalizes $\Gamma_0(N)$, so gives automorphism

$$Y_0(N) \xrightarrow{\sim} Y_0(N)$$

$$\Gamma_0(N)\tau \mapsto \Gamma_0(N) w_N(\tau).$$

i) Show that this is an involution

Pf: That is to show $\Gamma_0(N) w_N(w_N(\tau)) = \Gamma_0(N)\tau$.

$$w_N(\tau) = \frac{-1}{N\tau}. \quad w_N(w_N(\tau)) = \frac{-1}{N\frac{-1}{N\tau}} = \tau.$$

ii) describe the corresponding automorphism of the moduli space $S_0(N)$.

Sol:

$$\begin{array}{ccc} Y_0(N) & \xrightarrow{\sim} & Y_0(N) \\ \uparrow\wr & & \uparrow\wr \\ \langle 1/N + \Lambda_\tau \rangle \; S_0(N) & \xrightarrow{\sim} & S_0(N) \end{array}$$

$$[\mathbb{C}/\Lambda_\tau, \langle 1/N + \Lambda_\tau \rangle] \mapsto \Gamma_0(N)\tau \mapsto \Gamma_0(N) w_N(\tau) \mapsto [\mathbb{C}/\Lambda_{w_N(\tau)}, \langle 1/N + \Lambda_{w_N(\tau)} \rangle]$$

$$\underset{\sim}{m = j(w_N, \tau)} [\mathbb{C}/\Lambda_\tau, \langle m/N + \Lambda_\tau \rangle] \overset{m=}{\sim}$$

1.5.6 $N \in \mathbb{Z}_+$. p prime. $\Gamma^0(p) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & 0 \\ * & * \end{pmatrix} (p) \right\}$

and $\Gamma_1^0(N,p) = \Gamma_1(N) \cap \Gamma^0(p)$.

$Y_1^0(N,p) = Y(\Gamma_1^0(N,p))$.

An enhanced elliptic curve for $\Gamma_1^0(N,p)$ is an ordered triple (E, C, Q). $|C| = p$ $\operatorname{ord} Q = N$.

$C \cap \langle Q \rangle = \{0_E\}$.

$(E, C, Q) \sim (E', C', Q')$ if $\exists$ $E \xrightarrow{\sim} E'$

$C \to C'$ $Q \mapsto Q'$.

$S_1^0(N,p) = \{ (E, C, Q) \text{ for } \Gamma_1^0(N,p) \} / \sim$

$= \{ [E, C, Q] \text{ for } \Gamma_1^0(N,p) \}$

(a) By proof of Thm 1.5.11(b). $[E, C, Q] = [\mathbb{C}/\Lambda_{\tau'}, C, 1/N + \Lambda_{\tau'}]$

Show C has a generator of the form

$(\tau' + j)/p + \Lambda_{\tau'}$ or $1/p + \Lambda_{\tau'}$, the latter only if $p \nmid N$.

Pf: $\cancel{\Gamma_1^0(N,p)\tau \xrightarrow{\psi_0'}}$ $[\mathbb{C}/\Lambda_{\tau'}, C, 1/N + \Lambda_{\tau'}] \mapsto \Gamma_1^0(N,p)\tau'$.

$|C| = p$, so must be $z/p + \Lambda_{\tau'}$. $z \in \mathbb{C}$.

$C \cap \langle Q \rangle$